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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab experiment-5

Experiment 5: Missionaries and Cannibals Problem

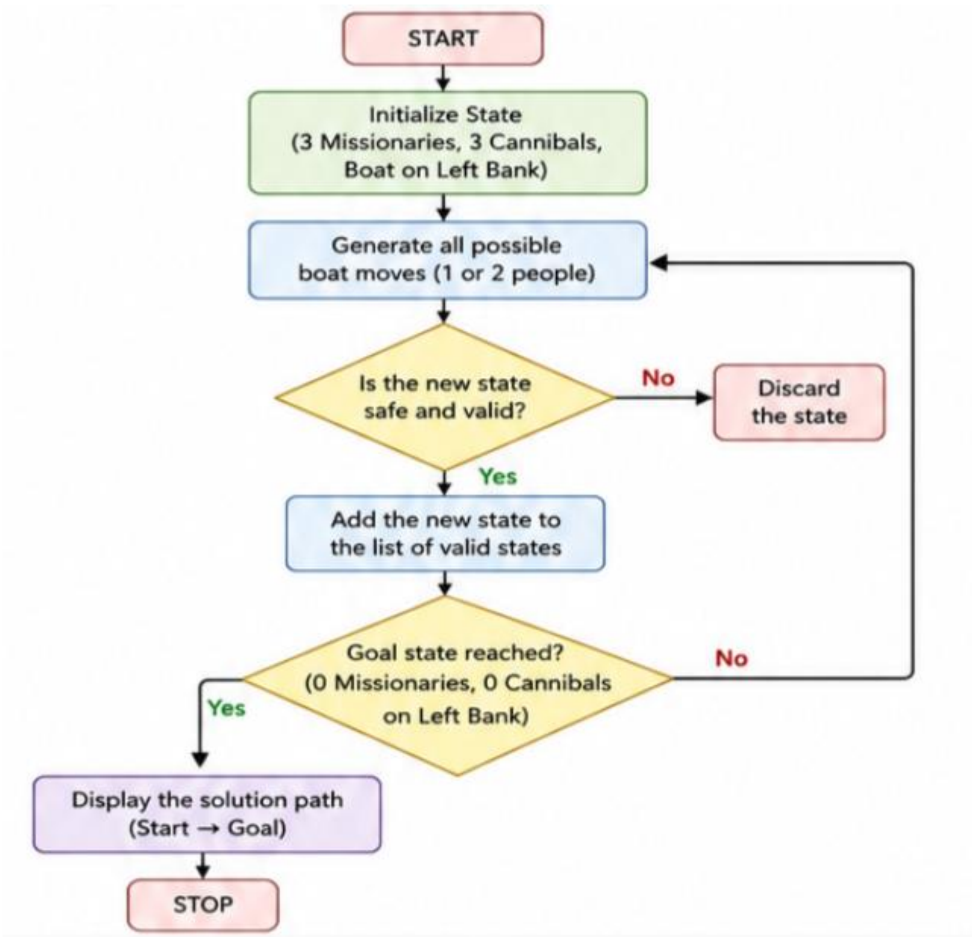
Aim

To develop a Python program to solve the Missionaries and Cannibals problem using State Space Search.

Objective

To transport three missionaries and three cannibals safely across the river without allowing cannibals to outnumber missionaries on either bank.

Flowchart



Algorithm

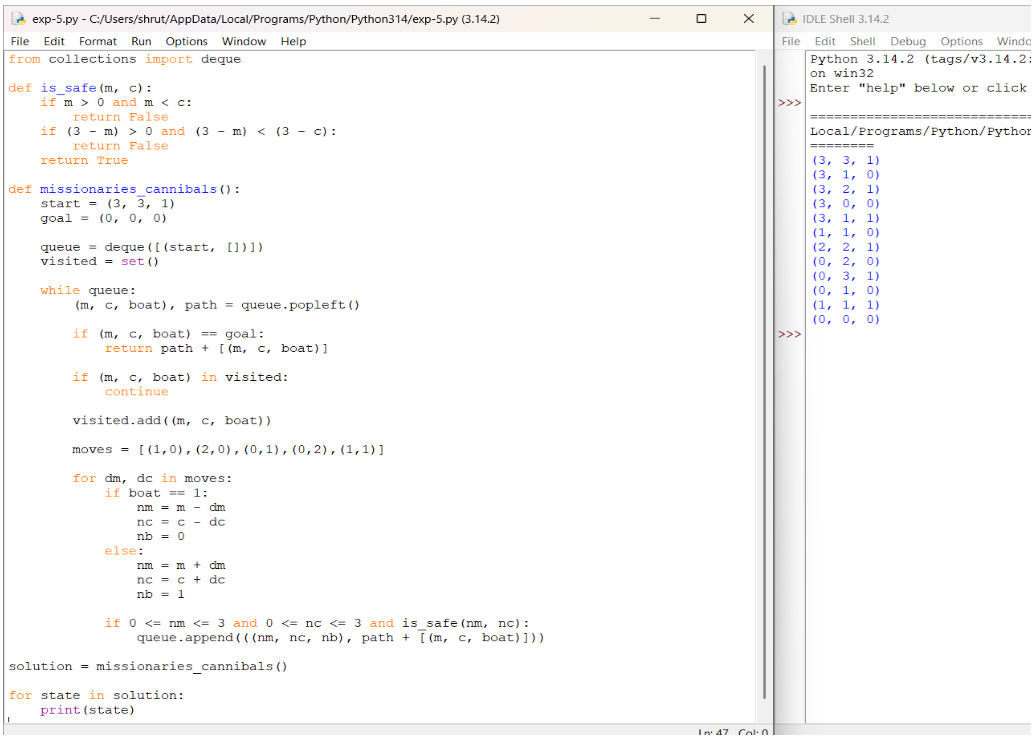
1. Initialize the starting state (3 missionaries, 3 cannibals, boat on left).
2. Generate all possible boat moves.
3. Check whether each new state is safe.
4. Ignore invalid states.
5. Continue exploring valid states until the goal state is reached.
6. Display the solution path.

Code

```
from collections import deque


def is_safe(m, c):
    if m > 0 and m < c:
        return False
    if (3 - m) > 0 and (3 - m) < (3 - c):
        return False
    return True
def missionaries_cannibals():
    start = (3, 3, 1)
    goal = (0, 0, 0)
    queue = deque([(start, [])])
    visited = set()
    while queue:
        (m, c, boat), path = queue.popleft()
        if (m, c, boat) == goal:
            return path + [(m, c, boat)]
```

```
        if (m, c, boat) in visited:
            continue
        visited.add((m, c, boat))
        moves = [(1,0),(2,0),(0,1),(0,2),(1,1)]
        for dm, dc in moves:
            if boat == 1:
                nm = m - dm
                nc = c - dc
                nb = 0
            else:
                nm = m + dm
                nc = c + dc
                nb = 1
            if 0 <= nm <= 3 and 0 <= nc <= 3 and is_safe(nm, nc):
                queue.append(((nm, nc, nb), path + [(m, c, boat)]))

solution = missionaries_cannibals()

for state in solution:
    print(state)
```

Output



Result

The Missionaries and Cannibals problem was successfully solved.

Conclusion

The program safely transported all missionaries and cannibals without violating the constraints.